

Exercise 3

chl

Exercise of Chapter 3 in the book. [Shakarchi, Stein] Complex Analysis (2003)

3-8: 4,5,6,8,9,10,11,12,13,14.

2026 年 3 月 20 日

4.

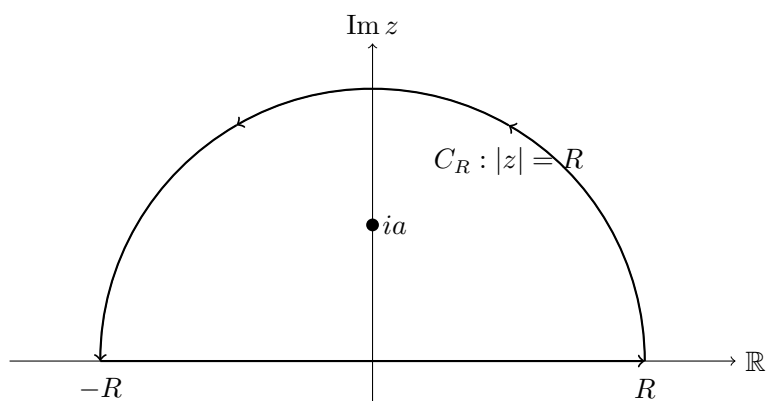
Show that

$$\int_{-\infty}^{\infty} \frac{x \sin x}{x^2 + a^2} dx = \pi e^{-a}, \quad \text{for all } a > 0.$$

Proof. Consider

$$2\pi i \operatorname{Res}_{ia} f(z) = \int_{\gamma} f(z) dz = \int_{C_R} f(z) dz + \int_{-R}^R f(z) dz \quad f(z) = \frac{ze^{iz}}{z^2 + a^2}$$

where :



Calculate the residue of f at ia :

$$\lim_{z \rightarrow ia} (z - ia) f(z) = \lim_{z \rightarrow ia} (z - ia) \frac{ze^{iz}}{(z - ia)(z + ia)} = \lim_{z \rightarrow ia} \frac{ze^{iz}}{z + ia} = \frac{iae^{-a}}{2ia} = \frac{e^{-a}}{2}$$

We next calculate the integral over the semicircle C_R . Let $a < R \rightarrow \infty$, then

$$\left| \int_{C_R} f(z) dz \right| \leq \int_{C_R} |f(z)| |dz| \leq \frac{R \cdot e^{-R \sin a}}{R^2 - a^2} \cdot \pi R \rightarrow \pi e^{-R \sin a} \rightarrow 0$$

Therefore,

$$\int_{\mathbb{R}} \frac{x e^{ix}}{x^2 + a^2} dx = 2\pi i \cdot \frac{e^{-a}}{2} = \pi i e^{-a}$$

Take the imaginary parts of the above identity, we get

$$\int_{-\infty}^{\infty} \frac{x \sin x}{x^2 + a^2} dx = \pi e^{-a}$$

□

5.

Use contour integration to show that

$$\int_{-\infty}^{\infty} \frac{e^{-2\pi i x \xi}}{(1 + x^2)^2} dx = \frac{\pi}{2} (1 + 2\pi |\xi|) e^{-2\pi |\xi|}$$

for all ξ real.

Proof. If $\xi \geq 0$, consider the contour in the lower half plane,

$$-2\pi i \operatorname{Res}_{-i} f = \int_{\gamma} f(z) dz = \int_{-R}^R f(x) dx + \int_{C_R} f(z) dz \quad f(z) = \frac{e^{-2\pi i z \xi}}{(1 + z^2)^2}.$$

Calculate the residue of f at $-i$. Since $-i$ is a pole of order 2, we have

$$\operatorname{Res}_{-i} f = \frac{d}{dz} \left(\frac{e^{-2\pi i \xi z}}{(z - i)^2} \right) \Big|_{z=-i}.$$

Let $g(z) = e^{-2\pi i \xi z}$, $h(z) = (z - i)^{-2}$. Then

$$g'(z) = -2\pi i \xi e^{-2\pi i \xi z}, \quad h'(z) = -2(z - i)^{-3}.$$

Thus

$$\begin{aligned}
\text{Res}_{-i} f &= g'(-i)h(-i) + g(-i)h'(-i) \\
&= (-2\pi i \xi e^{-2\pi \xi}) \frac{1}{(-2i)^2} + e^{-2\pi \xi} \left(-\frac{2}{(-2i)^3} \right) \\
&= \frac{\pi i \xi}{2} e^{-2\pi \xi} - \frac{i}{4} e^{-2\pi \xi} = i e^{-2\pi \xi} \left(\frac{\pi \xi}{2} - \frac{1}{4} \right).
\end{aligned}$$

For the integral over the semicircle C_R ,

$$\left| \int_{C_R} f(z) dz \right| \leq \int_{C_R} \frac{|e^{-2\pi i \xi z}|}{|1+z^2|^2} |dz| \leq \frac{\pi R}{(R^2-1)^2} \rightarrow 0.$$

Hence

$$\begin{aligned}
\int_{-\infty}^{\infty} \frac{e^{-2\pi i x \xi}}{(1+x^2)^2} dx &= -2\pi i \text{Res}_{-i} f \\
&= -2\pi i \cdot i e^{-2\pi \xi} \left(\frac{\pi \xi}{2} - \frac{1}{4} \right) \\
&= \frac{\pi}{2} (1 + 2\pi \xi) e^{-2\pi \xi}, \quad \xi \geq 0.
\end{aligned}$$

If $\xi < 0$, close in the upper half-plane. Similarly,

$$\int_{-\infty}^{\infty} \frac{e^{-2\pi i x \xi}}{(1+x^2)^2} dx = \frac{\pi}{2} (1 - 2\pi \xi) e^{2\pi \xi}.$$

Thus, for all $\xi \in \mathbb{R}$,

$$\int_{-\infty}^{\infty} \frac{e^{-2\pi i x \xi}}{(1+x^2)^2} dx = \frac{\pi}{2} (1 + 2\pi |\xi|) e^{-2\pi |\xi|}.$$

□

6.

Show that

$$\int_{-\infty}^{\infty} \frac{dx}{(1+x^2)^{n+1}} = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)} \cdot \pi.$$

Proof. Consider

$$2\pi i \operatorname{Res}_i f = \int_{\gamma} f(z) dz = \int_{C_R} f(z) dz + \int_{-R}^R f(x) dx \quad f(z) = (1+z^2)^{-n-1}$$

Calculate the residue of f at i . Since i is a pole of order $n+1$, we have

$$\begin{aligned} \operatorname{Res}_i f &= \frac{1}{n!} \lim_{z \rightarrow i} \frac{d^n}{dz^n} ((z-i)^{n+1} (1+z^2)^{-n-1}) = \frac{1}{n!} \lim_{z \rightarrow i} \frac{d^n}{dz^n} ((z+i)^{-n-1}) \\ &= \frac{1}{n!} (-1)^n \frac{(2n)!}{n!} (2i)^{-2n-1} = \frac{(2n)!}{(n!)^2 2^{2n}} \cdot \frac{1}{2i} = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)} \cdot \frac{1}{2i} \end{aligned}$$

For the integral over the semicircle C_R ,

$$\left| \int_{C_R} f(z) dz \right| \leq \int_{C_R} \frac{1}{|1+z^2|^{n+1}} |dz| \leq \frac{\pi R}{(R^2-1)^{n+1}} \rightarrow 0$$

Hence,

$$\int_{-\infty}^{\infty} \frac{dx}{(1+x^2)^{n+1}} = 2\pi i \operatorname{Res}_i f = 2\pi i \cdot \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)} \cdot \frac{1}{2i} = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)} \cdot \pi$$

□

8.

Prove that

$$\int_0^{2\pi} \frac{d\theta}{a + b \cos \theta} = \frac{2\pi}{\sqrt{a^2 - b^2}}$$

if $a > |b|$ and $a, b \in \mathbb{R}$.

Proof. Let $z = e^{i\theta}$, then $\cos \theta = \frac{z+z^{-1}}{2}$ and $d\theta = \frac{dz}{iz}$. Thus,

$$\int_0^{2\pi} \frac{d\theta}{a + b \cos \theta} = \int_{|z|=1} \frac{1}{a + b \frac{z+z^{-1}}{2}} \cdot \frac{dz}{iz} = 2i \int_{|z|=1} \frac{dz}{bz^2 + 2az + b}$$

The poles of the integrand are the roots of $bz^2 + 2az + b = 0$, i.e.

$$z_{1,2} = -\frac{a}{b} \pm \sqrt{\left(\frac{a}{b}\right)^2 - 1}$$

Since $a > |b|$, $\left(\frac{a}{b}\right)^2 > 1$, we know that $z_1 = -\frac{a}{b} + \sqrt{\left(\frac{a}{b}\right)^2 - 1}$ is inside the unit circle and z_2 is outside. Therefore, we only need to calculate the residue at z_1 :

$$\operatorname{Res}_{z_1} f = \operatorname{Res}_{z_1} \frac{2i}{bz^2 + 2az + b} = \operatorname{Res}_{z_1} \frac{2i}{b(z - z_1)(z - z_2)} = \frac{2i}{b(z_1 - z_2)}.$$

Note that

$$z_1 - z_2 = 2\sqrt{\left(\frac{a}{b}\right)^2 - 1} = 2\frac{\sqrt{a^2 - b^2}}{|b|}.$$

Then,

$$\operatorname{Res}_{z_1} f = \frac{i|b|}{b\sqrt{a^2 - b^2}} = \pm \frac{i}{\sqrt{a^2 - b^2}}.$$

By the residue theorem,

$$\int_0^{2\pi} \frac{d\theta}{a + b \cos \theta} = 2\pi i \operatorname{Res}_{z_1} f = 2\pi i \cdot \frac{\pm i}{\sqrt{a^2 - b^2}} = \frac{\mp 2\pi}{\sqrt{a^2 - b^2}}.$$

Since $a > |b| \implies a + b \cos \theta \geq a - b > 0$, the integrand is positive, we must have

$$\int_0^{2\pi} \frac{d\theta}{a + b \cos \theta} = \frac{2\pi}{\sqrt{a^2 - b^2}}.$$

□

9.

Show that

$$\int_0^1 \log(\sin \pi x) dx = -\log 2.$$

[Hint: Use the contour shown in Figure 9.]

Figure 9. Contour in Exercise 9

Proof. Note that

$$\sin \pi z = \frac{e^{i\pi z} - e^{-i\pi z}}{2i} = \frac{i}{2} e^{-i\pi z} (1 - e^{2\pi iz})$$

Then

$$-2ie^{i\pi z} \sin \pi z = 1 - e^{2\pi iz} \implies \log(-2ie^{i\pi z} \sin \pi z) = \log(1 - e^{2\pi iz}) \triangleq f(z)$$

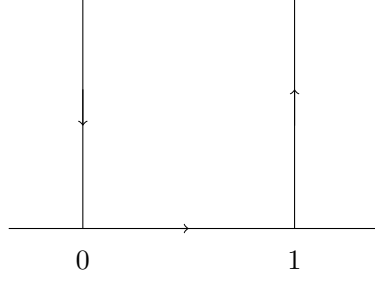


图 1: **Figure 9.** Contour in Exercise 9

Let $0 < R \rightarrow \infty$, $0 < \epsilon \rightarrow 0$, consider the contour $\gamma : \epsilon \rightarrow 1 - \epsilon \rightarrow 1 - \epsilon + iR \rightarrow \epsilon + iR \rightarrow \epsilon$. Note that $f(z+1) = f(z)$ and f has no singularities inside the contour, , we have

$$\begin{aligned}
 0 &= \int_{\gamma} f(z)dz = \int_{\epsilon}^{1-\epsilon} f(z)dz + \int_{1-\epsilon}^{1-\epsilon+iR} f(z)dz + \int_{1-\epsilon+iR}^{\epsilon+iR} f(z)dz + \int_{\epsilon+iR}^{\epsilon} f(z)dz \\
 &= \int_{\epsilon}^{1-\epsilon} f(x)dx + \int_{\epsilon}^{\epsilon+iR} f(z)dz - \int_{-\epsilon}^{-\epsilon+iR} f(z)dz - \int_{\epsilon}^{1-\epsilon} \log(1 - e^{2\pi i(x+iR)})dx \\
 &\rightarrow \int_{\epsilon}^{1-\epsilon} f(x)dx - \int_{\epsilon}^{1-\epsilon} \log(1 - e^{2\pi ix} e^{-2\pi R})dx \\
 &\rightarrow \int_{\epsilon}^{1-\epsilon} f(x)dx - \int_{\epsilon}^{1-\epsilon} \log(1)dx = \int_{\epsilon}^{1-\epsilon} \log(1 - e^{2\pi ix})dx
 \end{aligned}$$

Note that for $0 < x < 1$, we have $|-2ie^{i\pi x} \sin \pi x| = 2 \sin \pi x$, $\text{Arg}(-2ie^{i\pi x} \sin \pi x) = \pi(x - 1/2)$. Thus

$$\begin{aligned}
 0 &= \int_{\epsilon}^{1-\epsilon} \log(1 - e^{2\pi ix})dx = \int_{\epsilon}^{1-\epsilon} \log(-2ie^{i\pi x} \sin \pi x)dx \\
 &= \int_{\epsilon}^{1-\epsilon} \left(\log(2 \sin \pi x) + i\pi(x - \frac{1}{2}) \right) dx \\
 &\rightarrow \log 2 + \int_{\epsilon}^{1-\epsilon} \log(\sin \pi x)dx + i \int_{\epsilon}^{1-\epsilon} \pi(x - \frac{1}{2})dx \\
 &\rightarrow \log 2 + \int_{\epsilon}^{1-\epsilon} \log(\sin \pi x)dx
 \end{aligned}$$

Therefore, when $R \rightarrow \infty, \epsilon \rightarrow 0$, we have

$$\implies \int_0^1 \log(\sin \pi x) dx = -\log 2$$

□

10.

Show that if $a > 0$, then

$$\int_0^\infty \frac{\log x}{x^2 + a^2} dx = \frac{\pi}{2a} \log a.$$

[Hint: Use the contour in Figure 10.]

Figure 10. Contour in Exercise 10

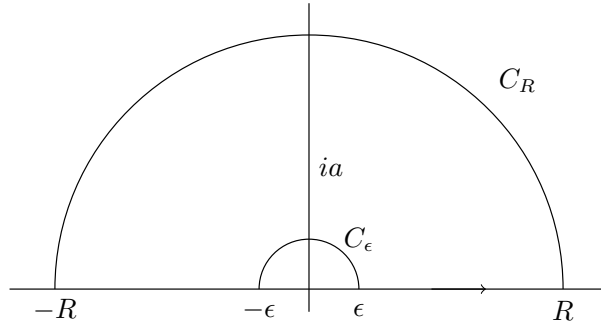


Figure 10. Contour in Exercise 10

Proof. We define $\log(\cdot)$ on $\mathbb{C} \setminus \{ti : t \leq 0\}$ $\log(re^{i\theta}) = \log r + i\theta, \theta \in (-\pi/2, 3\pi/2)$. Consider

$$2\pi i \operatorname{Res}_{ai} f = \int_{\gamma} f(z) dz \quad f(z) = \frac{\log z}{z^2 + a^2}$$

Calculate the residue of f at ai . Since ai is a simple pole, we have

$$\operatorname{Res}_{ai} f = \lim_{z \rightarrow ai} (z - ai) \frac{\log z}{z^2 + a^2} = \lim_{z \rightarrow ai} \frac{\log z}{z + ai} = \frac{\log(ai)}{2ai} = \frac{\log a + i\pi/2}{2ai}$$

For the integral over C_R , when $R \rightarrow \infty$, we have

$$\left| \int_{C_R} f(z) dz \right| \leq \int_{C_R} \frac{|\log z|}{|z^2 + a^2|} |dz| \leq \int_{C_R} \frac{\log R}{R^2 - a^2} |dz| = \frac{\log R}{R^2 - a^2} \cdot \pi R \rightarrow 0$$

For the integral over C_ϵ , when $\epsilon \rightarrow 0$, we have

$$\left| \int_{C_\epsilon} f(z) dz \right| \leq \int_{C_\epsilon} \frac{|\log z|}{|z^2 + a^2|} |dz| \leq \int_{C_\epsilon} \frac{|\log \epsilon|}{a^2 - \epsilon^2} |dz| = \frac{|\log \epsilon|}{a^2 - \epsilon^2} \cdot \pi \epsilon \rightarrow 0$$

Hence,

$$\int_0^\infty \frac{2 \log x + i\pi + i \cdot 0}{x^2 + a^2} dx = \int_{\gamma} f(z) dz = 2\pi i \operatorname{Res}_{ai} f = 2\pi i \cdot \frac{\log a + i\pi/2}{2ai} = \frac{\pi \log a}{a} + \frac{i\pi^2}{2a}$$

Take the real and imaginary parts of the above identity, we get

$$\int_0^\infty \frac{\log x}{x^2 + a^2} dx = \frac{\pi}{2a} \log a, \quad \int_0^\infty \frac{dx}{x^2 + a^2} = \frac{\pi}{2a}$$

□

11.

Show that if $|a| < 1$, then

$$\int_0^{2\pi} \log |1 - ae^{i\theta}| d\theta = 0.$$

Then, prove that the above result remains true if we assume only that $|a| \leq 1$.

Proof. $|a| < 1$: Let

$$\log(1 - az) = - \sum_{n=1}^{\infty} \frac{(az)^n}{n}, \quad |z| = 1.$$

Taking real parts on both sides, we have

$$\log |1 - ae^{i\theta}| = \operatorname{Re} \log(1 - ae^{i\theta}) = -\operatorname{Re} \sum_{n=1}^{\infty} \frac{a^n e^{in\theta}}{n}.$$

Hence

$$\int_0^{2\pi} \log |1 - ae^{i\theta}| d\theta = -\Re \sum_{n=1}^{\infty} \frac{a^n}{n} \int_0^{2\pi} e^{in\theta} d\theta = 0.$$

$|a| = 1$: Let $a = e^{i\phi}$, by the periodicity of the $z \mapsto e^{iz}$ with period 2π , we have

$$\int_0^{2\pi} \log |1 - ae^{i\theta}| d\theta = \int_0^{2\pi} \log |1 - e^{i(\theta+\phi)}| d\theta = \int_\phi^{2\pi+\phi} \log |1 - e^{i\theta}| d\theta = \int_0^{2\pi} \log |1 - e^{i\theta}| d\theta$$

We now suppose $a = 1$ without loss of generality, then

$$\begin{aligned} \int_0^{2\pi} \log |1 - e^{i\theta}| d\theta &= \int_0^{2\pi} \log 2 \left| \sin \frac{\theta}{2} \right| d\theta = \int_0^{2\pi} \log 2 d\theta + \int_0^{2\pi} \log \left| \sin \frac{\theta}{2} \right| d\theta \\ &= 2\pi \log 2 + 2 \int_0^\pi \log |\sin x| dx \\ &= 2\pi \log 2 + 2\pi \int_0^1 \log |\sin \pi x| dx \\ &= 2\pi \log 2 - 2\pi \log 2 = 0 \end{aligned}$$

□

12.

Suppose u is not an integer. Prove that

$$\sum_{n=-\infty}^{\infty} \frac{1}{(u+n)^2} = \frac{\pi^2}{(\sin \pi u)^2}$$

by integrating

$$f(z) = \frac{\pi \cot \pi z}{(u+z)^2}$$

over the circle $|z| = R_N = N + 1/2$ (N integral, $N \geq |u|$), adding the residues of f inside the circle, and letting N tend to infinity.

Note. Two other derivations of this identity, using Fourier series, were given in Book I.

Proof. Consider

$$\int_{\gamma} f(z) dz \quad f(z) = \frac{\pi \cot \pi z}{(u+z)^2} \quad \gamma : |z| = R_N = N + 1/2$$

All the poles of f inside the circle are $z = -u$ (order 2), $0, \pm 1, \pm 2, \dots, \pm N$ (order 1). Calculate the residues of f at these poles:

$$Res_{-u} f = \lim_{z \rightarrow -u} \frac{d}{dz} \left((z+u)^2 \cdot \frac{\pi \cot \pi z}{(u+z)^2} \right) = \lim_{z \rightarrow -u} \frac{d}{dz} (\pi \cot \pi z) = -\pi^2 \csc^2 \pi u$$

$$Res_k f = \lim_{z \rightarrow k} (z-k) \cdot \frac{\pi \cot \pi z}{(u+z)^2} = \lim_{z \rightarrow k} (z-k) \cdot \frac{\pi \cos \pi z}{\sin \pi z} \cdot \frac{1}{(u+z)^2} = \frac{1}{(u+k)^2}$$

where $k = 0, \pm 1, \pm 2, \dots, \pm N$.

Therefore, by the residue theorem,

$$\int_{\gamma} f(z) dz = 2\pi i \left(-\pi^2 \csc^2 \pi u + \sum_{n=-N}^N \frac{\cos n\pi}{(u+n)^2} \right) = 2\pi i \left(-\frac{\pi^2}{(\sin \pi u)^2} + \sum_{n=-N}^N \frac{1}{(u+n)^2} \right)$$

We know that $\exists C > 0$ s.t. $\forall |u| < N \in \mathbb{N}$, $|\pi \cot \pi z| \leq C$ for all z on the circle $|z| = R_N$. Thus,

$$\left| \int_{\gamma} f(z) dz \right| \leq \int_{\gamma} \frac{|\pi \cot \pi z|}{|u+z|^2} |dz| \leq \int_{\gamma} \frac{C}{(N+1/2-|u|)^2} |dz| = \frac{C \cdot 2\pi(N+1/2)}{(N+1/2-|u|)^2} \rightarrow 0$$

This gives

$$\sum_{n=-\infty}^{\infty} \frac{1}{(u+n)^2} = \frac{\pi^2}{(\sin \pi u)^2}$$

□

13.

Suppose $f(z)$ is holomorphic in a punctured disc $D_r(z_0) - \{z_0\}$. Suppose also that

$$|f(z)| \leq A|z - z_0|^{-1+\epsilon}$$

for some $\epsilon > 0$, and all z near z_0 . Show that the singularity of f at z_0 is removable.

Proof. Since $f(z)$ is holomorphic in a punctured disc $D_r(z_0) - \{z_0\}$, we can write $f(z)$ as a Laurent series around z_0 :

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n$$

For $m \geq 1$, we have

$$a_{-m} = \frac{1}{2\pi i} \int_{|z-z_0|=r} \frac{f(z)}{(z - z_0)^{-m+1}} dz$$

Note that $\epsilon + m - 1 \geq \epsilon > 0$, when $0 < \rho \rightarrow 0$,

$$|a_{-m}| \leq \frac{1}{2\pi} \int_{|z-z_0|=\rho} \frac{|f(z)|}{|z - z_0|^{-m+1}} |dz| \leq \frac{1}{2\pi} \int_{|z-z_0|=\rho} A|z - z_0|^{-1+\epsilon+m-1} |dz| = A\rho^{\epsilon+m-1} \rightarrow 0$$

Therefore,

$$f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n$$

We can then define $f(z_0) = a_0$, and f is holomorphic in $D_r(z_0)$, which means the singularity of f at z_0 is removable. □

14.

Prove that all entire functions that are also injective take the form

$$f(z) = az + b \quad \text{with } a, b \in \mathbb{C}, \text{ and } a \neq 0.$$

[Hint: Apply the Casorati-Weierstrass theorem to $f(1/z)$.]

Proof. Consider $g(z) = f(1/z)$, then g is holomorphic in $\mathbb{C} \setminus \{0\}$. If 0 is an essential singularity of g , then $\forall r > 0$, $g(\{0 < |z| < r\})$ is dense in $\mathbb{C}$ (Casorati-Weierstrass).

Fix $z_1 \neq 0$ and set $w_1 = g(z_1)$, then $\exists z_n \rightarrow 0$ s.t. $g(z_n) \rightarrow w_1$. Let $h(z) = g(z) - w_1$, then h is holomorphic on $0 < |z| < r$, and $h(z_1) = 0$. Since $h \not\equiv 0$, the zero z_1 is isolated, $\exists$ a closed disc $\overline{D}$ centered at z_1 s.t. z_1 is the only zero of h in $\overline{D}$.

$\forall n$, define

$$h_n(z) = g(z) - g(z_n) = h(z) - h(z_n).$$

Then $h_n \rightarrow h$ uniformly on $\overline{D}$. By Hurwitz's theorem, for all sufficiently large n , each h_n has a zero $\zeta_n \in D$. Thus

$$g(\zeta_n) = g(z_n).$$

Since $z_n \rightarrow 0$ while D is centered at $z_1 \neq 0$, we have $\zeta_n \neq z_n$ for all large n , contradicting the injectivity of g .

If 0 is a removable singularity of g , then we can define $g(0) = c$ for some $c \in \mathbb{C}$. This means $\lim_{|z| \rightarrow \infty} f(z) = c \implies f$ is bounded in $\mathbb{C} \implies f$ is constant (Liouville), which contradicts the injectivity of f .

If 0 is a pole of g , then we can write $g(z) = \sum_{n=-m}^{\infty} a_n z^n$ for some $m \geq 1$ and $a_{-m} \neq 0$. Then

$$f(z) = g(1/z) = \sum_{n=-m}^{\infty} a_n z^{-n} = a_{-m} z^m + \sum_{n=-\infty}^{m-1} b_n z^n$$

Since f is an entire function, we have

$$f(z) = \sum_{n=0}^{\infty} b_n z^n \quad f'(z) = \sum_{n=1}^{\infty} n b_n z^{n-1}$$

$\exists z_0 \in \mathbb{C}$ s.t. $f'(z_0) = 0$, then we have

$$f(z) = f(z_0) + c_k(z - z_0)^k + \cdots = f(z_0) + c_k(z - z_0)^k(1 + o(1)) \quad k \geq 2, c_k \neq 0$$

Consider $w \rightarrow f(z_0)$, the equation $f(z) = w \iff c_k(z - z_0)^k(1 + o(1)) = w - f(z_0)$ has k solutions near z_0 , which contradicts the injectivity of f .

Therefore, $m \leq 1$, together with $m > 0$ (*injectivity*), $m = 1$, i.e.

$$f(z) = az + b \quad \text{with } a, b \in \mathbb{C}, \text{ and } a \neq 0.$$

□